Endsem **EE1101 Signals and Systems** Date: 03 May 2019

Name: **Roll No.**:

Time allotted: **3 hours** Maximum Marks: **80**

Instructions:

Answers to Qns 1-8 must be written in the space provided in the question paper.
Answers to Qns 9-13 must be written in additional sheets.

Indicate your instructor.

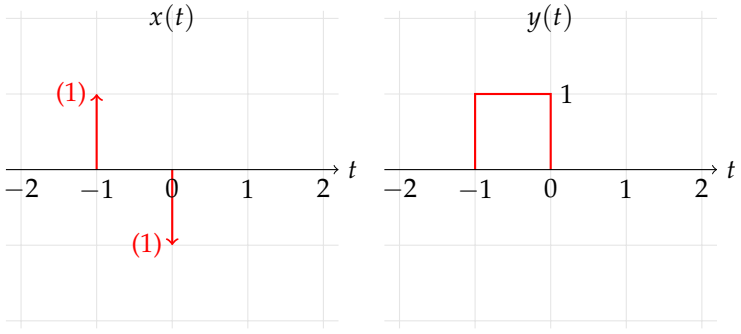
☐ Deepa V. ☐ Pradeep S. ☐ Shanti B. ☐ Venkatesh R.

Question:	1	2	3	4	5	6	7	8	9	10	11	12	13	Total
Points:	2	2	2	2	2	2	4	4	12	12	12	12	12	80
Score:														

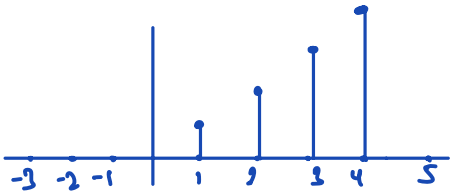
1. (2 points) Let $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$ where $x(t) = \delta(t+1) - 3\,\delta(3t)$. Plot $x(t)$ and $y(t)$.

u(t+1) - u(3t)

Solution:



1+1 marks,



$n<0 \Rightarrow x[n]=0$

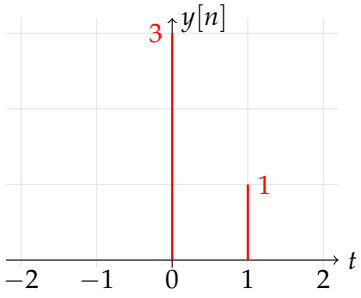
$n>4 \Rightarrow x[n]=0$

$x\in[0,4] \Rightarrow x[n]=n$

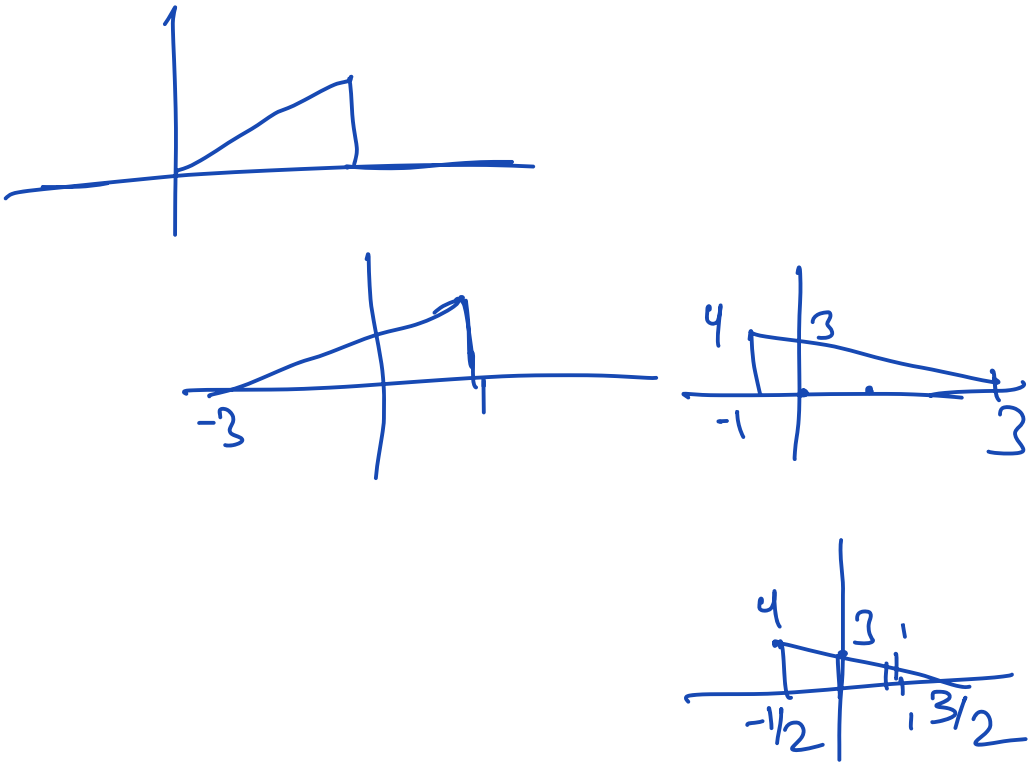
2. (2 points) Sketch $y[n] = x[-2n+3]$ where $x[n] = \begin{cases} nu[n] & n \le 4 \\ 0 & \text{elsewhere} \end{cases}$

$x[-2n+3] \rightarrow$ move left by 3, flip, squish by 2

Solution: we need $0 \le -2n+3 \le 4 \Rightarrow n \ge -1/2$ and $n \le 3/2 \Rightarrow 0 \le n \le 1$. So $y[0] = x[3] = 3$ and $y[1] = x[1] = 1$.



2 marks for plot, 1 mark partial credit if plot incorrect



3. (2 points) A signal whose frequency spectrum shown in Fig 1, is sampled with a unit impulse train with $\omega_s = 4\ rad/s$. Sketch the frequency spectrum of the sampled signal; indicate the values for $|\omega| \leq 10$.

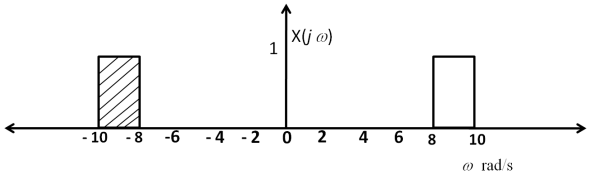
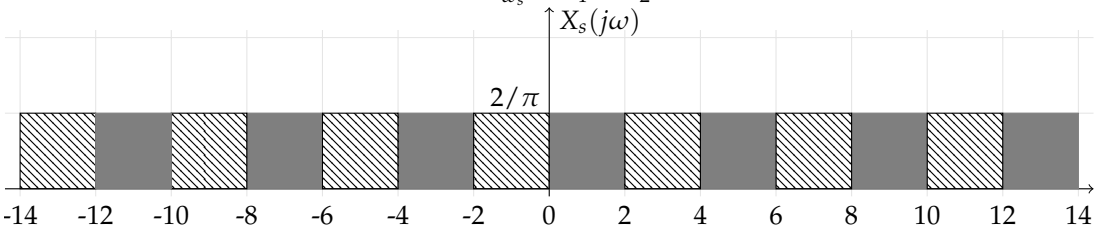


Figure 1: Frequency spectrum for Qn 3

Solution:

$$X_s(\omega) = \frac{1}{T}\sum_n X(j(\omega - n\omega_s))$$

$$T = \frac{2\pi}{\omega_s} = \frac{2\pi}{4} = \frac{\pi}{2}$$



2 marks, plot missing $1/T$ only one mark.

Sampling in Frequency Domain

Sampling a signal in time by an impulse train of frequency ω_s causes its spectrum to repeat (replicate) at intervals of ω_s in the frequency domain.

So, the spectrum after sampling, $X_s(j\omega)$, will be:

- The original spectrum, plus shifted copies every $\pm k\omega_s$, where k is an integer.

$\text{rect}(t/T) \leftrightarrow \frac{2T\sin(\omega T)}{\omega T}$

4. (2 points) Let $x(t) = \frac{sin(5\pi t)}{5\pi t}$. Find the value of $\int_{-\infty}^{\infty} x(t)^2\,dt$

Solution: If $x(t) \leftrightarrow X(j\omega)$, then by duality we have $X(t) \leftrightarrow 2\pi x(-j\omega)$.
Now $\text{rect}(\frac{t}{T}) \leftrightarrow \frac{2T\sin\omega T}{\omega T}$, therefore

$$\begin{aligned}
\frac{2W\sin tW}{tW} &\leftrightarrow 2\pi\text{rect}(\frac{\omega}{W}) \\
\Rightarrow \frac{\sin tW}{tW} &\leftrightarrow \frac{2\pi}{2W}\text{rect}(\frac{\omega}{W})|_{W=5\pi} = \frac{1}{5}\text{rect}(\frac{\omega}{5\pi})[1]\text{1 mark correct } X(j\omega) \\
\int_{-\infty}^{\infty} x(t)^2\,dt &= \frac{1}{2\pi}\int_{-\infty}^{\infty}|X(j\omega)|^2 d\omega = \frac{1}{2\pi}2(5\pi)\frac{1}{5^2} = \frac{1}{5}\text{1 mark}
\end{aligned}$$

Alternate solution, with convolution of the rect in the frequency domain w/o Parseval's theorem also ok. 1 mark for correct answer

5. (2 points) For a signal which is imaginary and even in the time domain, the Fourier transform is .....imaginary.....(real/imaginary) and .....even.....(even/odd). Justify your answer.

Solution: A real even signal ⇒ FT is real and even. 1 mark

Multiplying by a imaginary constant we obtain that the FT is imaginary and even. 1 mark

6. (2 points) The step response of an LTI system is $\cos(\omega_o t)u(t)$, find its impulse response.

$\delta(t) - u(t).\omega_0 \sin(\omega_0 t)$

Solution:

$$\begin{aligned} h(t) &= \frac{d}{dt}(s(t)) \text{1 mark} \\ &= (\frac{d}{dt}\cos(\omega_o t))u(t) + \cos(\omega_o t)\frac{d}{dt}u(t) \\ &= -\omega_o \sin(\omega_o t)u(t) + \cos(\omega_o t)\delta(t) \\ &= -\omega_o \sin(\omega_o t)u(t) + \delta(t) \text{1 mark} \end{aligned}$$

7. (4 points) The transfer function of an LTI system is given below. Write down the conditions on a, b, c so that the system is real, and stable and causal. Assume a, b, c are distinct.

$$H(s) = \frac{s+a}{(s+b)(s+c)}$$

Solution: System is real implies $H^*(s) = H(s^*)$ ie $H^*(s^*) = H(s)$, so the poles/zeros are either real or occur as conjugate pairs. 1 mark

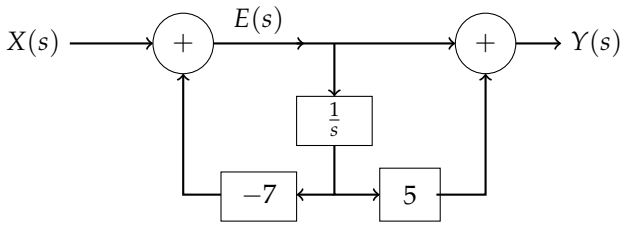
Therefore a is real and either b, c are real or complex conjugates $b = c^*$. 1 mark

System is casual implies that the ROC is a right half plane. 1 mark

System is stable implies that the $j\omega$ axis is to the right of the rightmost pole.

$Re\{b\}, Re\{c\} < 0$ 1 mark

8. (4 points) Find the transfer functions $\frac{E(s)}{X(s)}$ and $\frac{Y(s)}{X(s)}$.



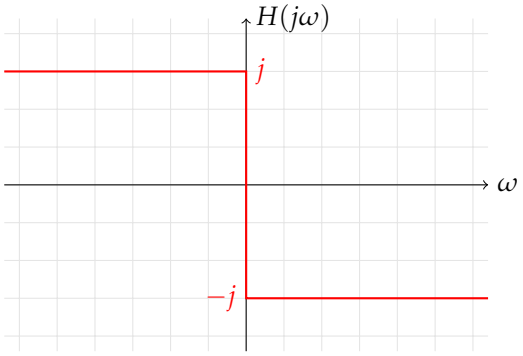
Solution:

$$
\begin{aligned}
E(s) &= X(s) - \frac{7E(s)}{s} \Rightarrow E(s) 1mark \\
&= \frac{1}{1+\frac{7}{s}} X(s) 1mark \\
Y(s) &= E(s) + \frac{5E(s)}{s} 1mark \\
\Rightarrow Y(s) &= (1+\frac{5}{s})E(s) = \frac{1+\frac{5}{s}}{1+\frac{7}{s}} X(s) 1mark
\end{aligned}
$$

9. (12 points) A signal $x(t)$ is passed through an ideal s shifter which is an LTI system (denoted as system H). Its frequency response is given as $H(j\omega) = \begin{cases} e^{\frac{-j\pi}{2}} & \omega > 0 \\ 0 & \omega = 0 \\ e^{\frac{j\pi}{2}} & \omega < 0 \end{cases}$

(a) Sketch $H(j\omega)$ [2]

Solution:



2 marks binary?

Should give credit for someone who plots mag and phase resp.

(b) Find the impulse response of this ideal phase shifter. [2]

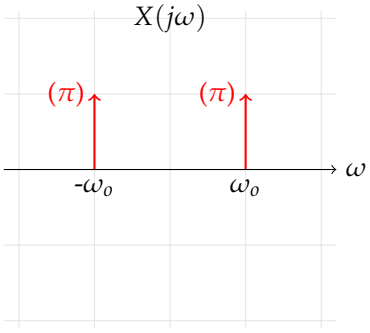
Solution: By duality $X(t) \leftrightarrow 2\pi x(-\omega)$ so $x(-\omega) \leftrightarrow \frac{X(t)}{2\pi}$ 1 mark

Since $\text{sign}(t) \leftrightarrow \frac{2}{j\omega}$ we have $j\text{sign}(-\omega) \leftrightarrow j\frac{1}{2\pi}\frac{2}{jt} = \frac{1}{\pi t}$ 1 mark

(c) Given $x(t) = cos(\omega_0 t)$, plot $X(j\omega)$. [2]

Solution:

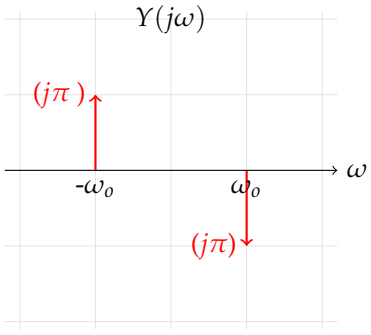
$\cos\omega_o t = \frac{e^{j\omega_0 t}+e^{-j\omega_0 t}}{2} \leftrightarrow \pi(\delta(\omega+\omega_0)+\delta(\omega-\omega_0))$ 1 mark



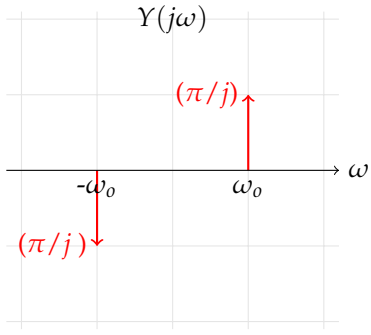
1 mark for plot 2 marks if plot is correct

(d) $x(t)$ given above is passed through H, to give an output $y(t)$. Find $Y(j\omega)$, $y(t)$; plot $Y(j\omega)$ [3]

Solution:



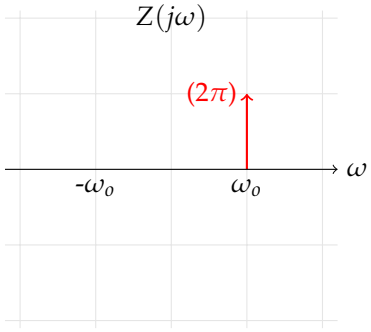
or



$Y(j\omega) \leftrightarrow \sin(\omega_o t)$ 1masrk $Y(j\omega)$, 1 mark $y(t)$

(e) Given $z(t) = x(t) + jy(t)$. Find $z(t), Z(j\omega)$; plot $Z(j\omega)$ [3]

Solution: $Z(j\omega) = X(j\omega) + jY(j\omega) = \delta(\omega - \omega_o) \leftrightarrow e^{j\omega_o t}$ 1 mark $z(t)$, + 1 mark $Z(j\omega)$



1 mark plot

10. (12 points) A causal LTI system is described by the following differential equation.

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = 2\frac{dx(t)}{dt} + x(t)$$

(a) Find the output $y(t)$ of the system for $x(t) = u(t)$ when the initial conditions are zero. [5]

Solution: Taking Laplace transforms we obtain

$$s^2Y(s) + 4sY(s) + 3Y(s) = (2s+1)X(s) = \frac{2s+1}{s} \quad 1mark$$

$$Y(s) = \frac{2s+1}{s(s+1)(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3} \quad 1mark$$

$$A = Y(s)s|_{s=0} = \frac{1}{3} \quad B = Y(s)(s+1)|_{s=-1} = \frac{1}{2} \quad C = Y(s)(s+3)|_{s=-3} = -\frac{5}{6} \quad 1mark$$

Taking inverse Laplace transform we obtain $y(t) = \frac{1}{3}u(t) + \frac{1}{2}e^{-t}u(t) - \frac{5}{6}e^{-3t}u(t)$ 1+1 mark

(b) Find the output $y(t)$ when the initial conditions are $y(0^-) = 1$ and $y'(0^-) = 2$ and $x(t) = 0$. [5]

Solution: Taking Laplace transforms we obtain

$$s^2Y(s) - sy(0^-) - y'(0^-) + 4sY(s) - 4y(0^-) + 3Y(s) = 0 \quad 1mark$$

$$s^2Y(s) + 4sY(s) + 3Y(s) = sy(0^-) + y'(0^-) + 4y(0^-) = s + 2 + 4 \quad 1mark$$

$$Y(s) = \frac{s+6}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3} \quad 1mark$$

$$A = Y(s)(s+1)|_{s=-1} = \frac{5}{2} \quad B = Y(s)(s+3)|_{s=-3} = -\frac{3}{2} \quad 1mark$$

Taking inverse Laplace transform we obtain $y(t) = \frac{5}{2}e^{-t}u(t) - \frac{3}{2}e^{-3t}u(t)$ 1 mark

(c) Total response of the system for input $x(t) = u(t)$ and initial conditions as in (b). [2]

Solution:

$$y(t) = \frac{1}{3}u(t) + \frac{1}{2}e^{-t}u(t) - \frac{5}{6}e^{-3t}u(t) + \frac{5}{2}e^{-t}u(t) - \frac{3}{2}e^{-3t}u(t) \quad 1mark$$

$$\Rightarrow y(t) = \frac{1}{3}u(t) + 3e^{-t}u(t) - \frac{7}{3}e^{-3t}u(t) \quad 1mark$$

11. (12 points) Find the Laplace transform of a causal signal $x(t)$ given below. Specify its ROC. State final value theorem and use it to find $x(\infty)$ [6+3+3]

$$x(t) = \begin{cases} t & 0 \le t < 2 \\ 2e^{-(t-2)} & 2 < t \end{cases}$$

Solution: We can write $x(t) = \underbrace{tu(t) - (t-2)u(t-2) - 2u(t-2)}_{x_1(t)} + \underbrace{2e^{-(t-2)}u(t-2)}_{x_2(t)}$ 1+1 mark

$$u(t) \leftrightarrow \frac{1}{s}; \quad tu(t) \leftrightarrow \frac{1}{s^2}; \quad e^{-t}u(t) \leftrightarrow \frac{1}{s+1} \; 1 + 1 mark$$

By linearity and time shifting properties, we have $X(s) = \frac{1}{s^2} - \frac{e^{-2s}}{s^2} - 2\frac{e^{-2s}}{s} + 2\frac{e^{-2s}}{s+1}$ 1+1 mark

ROC is right sided since the signal is causal. 1 mark

Note that $x_1(t)$ has ROC everywhere and $x_2(t)$ has ROC $\text{Re}\{s\} > -1$. 1 mark

Therefore the ROC of $x(t)$ is $\text{Re}\{s\} > -1$. 1 mark
(Note that pole/zero cancellation occurs at $s = 0$.)

The FVT states that

$$\begin{aligned} \lim_{t\to\infty} x(t) &= \lim_{s\to 0} sX(s) \; 1 mark \\ &= \lim_{s\to 0} \frac{1 - e^{-2s}}{s} - 2e^{-s} + \frac{se^{-2s}}{s+1} \; 1 mark \\ &= \lim_{s\to 0} \frac{2e^{-2s}}{1} - 2e^{-s} + \frac{se^{-2s}}{s+1} = 0 \; 1 \text{ mark} \end{aligned}$$

Solution:

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t)e^{-st}dt \; 1 mark \\ &= \int_0^{\infty} x(t)e^{-st} = \int_0^2 te^{-st} + \int_2^{\infty} e^{-(t-2)}e^{-st} \; 1 mark \\ &= \frac{te^{-st}}{-s}\Big|_0^2 - \int_0^2 \frac{e^{-st}}{-s}dt + e^{-(t-2)}e^{-st}\Big|_2^{\infty} \\ &= -\frac{2e^{-st}}{s} - \frac{e^{-st}}{s^2}\Big|_0^2 + e^2\frac{e^{-(s+1)t}}{-(s+1)}\Big|_2^{\infty} \\ &= -\frac{2e^{-st}}{s} - \frac{e^{-2s}}{s^2} + \frac{1}{s^2} + \frac{e^{-2s}}{s+1} \; 1 \text{ mark for each term} \end{aligned}$$

ROC Observe that pole/zero cancellation occurs for $s = 0$. Only one pole at $s = -1$. 1 mark
Rightsided ROC 1 mark
ROC = $\text{Re}\{s\} > -1$ 1 mark

12. (12 points) Find the exponential Fourier series coefficients of the function shown in Fig. 2. This signal is passed through an ideal low pass filter of cut off frequency 400 Hz. Find the output time domain signal. [7+5]

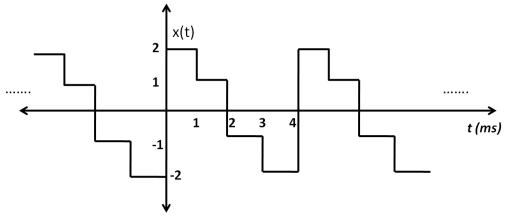


Figure 2: Time domain signal for Qn 13

Solution:

(a) Part a

(i) $T_0 = \frac{1}{4}ms; f_0 = 250Hz; \omega_0 = \frac{\pi}{2}krad/s$
Identification of T_o and ω_o 1 mark

(ii) $a_0 = 0$ 1 mark

(iii) Fourier series formula 1 mark

(iv) Integration substitution 2 marks

$$a_k = \frac{1}{T_0}\left[\int_0^{1ms} 2e^{-jk\omega_0 t}dt + \int_{1ms}^{2ms} 1e^{-jk\omega_0 t}dt + \int_{2ms}^{3ms} -1e^{-jk\omega_0 t}dt + \int_{3ms}^{4ms} -2e^{-jk\omega_0 t}dt\right]$$

(v) Correct answer 2 marks

Simplifying, $a_k = \frac{1}{jk2\pi}\left[2 - e^{-jk\frac{\pi}{2}} - 2e^{-2jk\frac{\pi}{2}} - e^{-3jk\frac{\pi}{2}} + 2e^{-4jk\frac{\pi}{2}}\right]$

(b) Part b

(i) Identification of correct harmonics in output 2 marks

(ii) Correct output 3 marks

Filter has a cut-off frequency : $400Hz$; Frequency of the signal : $250Hz$. Therefore, the dc component and first harmonic will be present at the output, $y(t)$.

$a_1 = \frac{3}{j\pi}$

$a_{-1} = \frac{-3}{j\pi}$

$y(t) = a_1 e^{j\omega_0 t} + a_{-1}e^{-j\omega_0 t} = \frac{6}{\pi}sin(\omega_o t)$

13. (12 points) Given $x(t) = \left(\frac{sin(80\pi t)}{\pi t}\right)^3$.
    (a) Find the Fourier transform of $x(t)$. [6]
    (b) Plot $X(j\omega)$. [3]
    (c) $x(t)$ is sampled with an impulse train with sampling frequency ω_s. Find the range of values of ω_s that satisfy Nyquist criterion (ie there is no aliasing). [3]

Solution:

(a) Part a

(i) Finding FT of $y(t) = \left(\frac{sin(80\pi t)}{\pi t}\right)$ 1 mark

$$Y(j\omega) = \begin{cases} 1 & |\omega| < W = 80\pi \\ 0 & elsewhere \end{cases}$$

(ii) Recognizing $x(t)^3 \leftrightarrow Y(j\omega) * Y(j\omega) * Y(j\omega)$ 1 mark
Let $y(t) = \left(\frac{sin(80\pi t)}{\pi t}\right)$

If $x(t) \rightarrow X(j\omega)$ and $y(t) \rightarrow Y(j\omega)$

then $X(j\omega) = \frac{1}{4\pi^2}\Big[Y(j\omega) \star Y(j\omega) \star Y(j\omega)\Big]$

(iii) Finding $Y(j\omega) * Y(j\omega)$ 2 marks

$$Y(j\omega) \star Y(j\omega) = \begin{cases} 0 & \omega \leq -2W \\ \omega + 2W & -2W < \omega \leq 0 \\ 2W - \omega & 0 < \omega \leq 2W \\ 0 & \omega > 2W \end{cases}$$

(iv) Complete convolution 2 marks

$$Y(j\omega) \star Y(j\omega) \star Y(j\omega) = \begin{cases} 0 & \omega \leq -3W \\ \int_{-2W}^{\omega+W} (\nu + 2W)d\nu = \left[2W^2 + 2W(\omega + W) + \frac{(\omega+W)^2}{2}\right] & -3W < \omega \leq \\ \int_{\omega-W}^{0} (\nu + 2W)d\nu + \int_{0}^{\omega+W} (2W - \nu)d\nu = \left[4W^2 - \frac{(\omega+W)^2}{2} - \frac{(\omega-W)^2}{2}\right] & -W < \omega \leq W \\ \int_{\omega-W}^{2W} (2W - \nu)d\nu = \left[2W^2 - 2W(\omega - W) + \frac{(\omega-W)^2}{2}\right] & W < \omega \leq 3W \\ 0 & \omega > 3W \end{cases}$$

Therefore,

$$X(j\omega) = \frac{1}{4\pi^2}\begin{cases} 0 & \omega \leq -3W \\ 2W^2 + 2W(\omega + W) + \frac{(\omega+W)^2}{2} & -3W < \omega \leq -W \\ 4W^2 - \frac{(\omega+W)^2}{2} - \frac{(\omega-W)^2}{2} & -W < \omega \leq W \\ 2W^2 - 2W(\omega - W) + \frac{(\omega-W)^2}{2} & W < \omega \leq 3W \\ 0 & \omega > 3W \end{cases}$$

(b) Part b

(i) Labelling of x-axis critical points 2 marks

(ii) Labelling of y-axis critical points 1 mark
Critical points $(-3W,0)$, $(-W, 2W^2)$, $(0, 3W^2)$, $(W, 2W^2)$, $(3W,0)$, with $W = 80\pi$.

(c) $x(t)$ is present from $-3W = 120$ to $3W = 120$

(i) Nyquist criterion 2 marks

(ii) Correct value of ω_s 1 mark
Largest frequency in $X(j\omega) = 3W = 240\pi$. Therefore $\omega_s > 480\pi$ to satisfy Nyquist criterion.

End Semester Examination - Solution Key
EE1101 Signals and Systems - Jan-May 2018

Duration:3 Hrs **Max Marks : 80**

Instructions

- ***Qn 1-10** : **2 marks each**. Answer briefly, with appropriate justification/counter example(s).*
- ***Qn 11-17** - Marks are shown in brackets. Show all working steps neatly.*
- *Do not write anything in the question paper. Attach the question paper with your answersheet.*

1. Does a two-sided signal exist such that the RoC of its Laplace Transform is a left half-plane ?

   Solution: Yes, for example $x(t) = u(-t) + e^{-t^2}$. Binary grading, full credit only if the example is correct.

2. Find the RoC of the Laplace transform of the signal $x(t) = \cos \pi t$.

   Solution: RoC is empty because the LT integral does not converge absolutely for any $s \in \mathbb{C}$. Binary grading.

3. When a causal LTI system is cascaded with a non-causal LTI system, then the resultant system must be non-causal. Is this statement true or false ?

   Solution: FALSE : When a LTI causal system is cascaded with a non-causal system, then resultant system need NOT be non-causal. Consider $y_1(t) = x(t-1)$ which is causal and $y_2(t) = x(t+1)$ which is non-causal. The resultant system is $y(t) = x(t)$, which is causal.

   Binary grading; credit for correct explantion.

4. For the LTI systems shown in Fig.1a., $y_1[n] = y_2[n]$. Is this statement true or false ?

   Solution : True; convolution property. Binary grading.

5. Find whether pole-zero plot shown in Fig. 1b could correspond to an even function in time ? If yes- find the required ROC. (Given that the transfer function is rational.)

   Solution: Given that $H(s)$ is rational and the zeros are at $\pm j$ and the poles are at ± 1; let $H(s) = \frac{A(s-j)(s+j)}{(s-1)(s+1)}$. Now, it is evident that $H(-s) = H(s)$. Thus, $H(s)$ is even. Since $H(s)$ is even, $h(t)$ can be even. However, if $h(t)$ is even, it should be either two sided or it has to be bounded and symmetric. For these to be satisfied, the ROC has to be a strip or the entire s-place. Since there are poles in the ROC, the ROC has to be a strip, defined by $-1 < Re\{s\} < 1$.

   1 mark for identifying $H(s)$ is even, and 1 mark for identifying the ROC ; marks to be given only if the supporting arguments are correct.

6. Sketch the magintude and phase response of a system that performs differentiation.

   Solution : See figure; 1 mark for the correct magnitude response, 1 mark for phase response - axes should be marked for full credit.

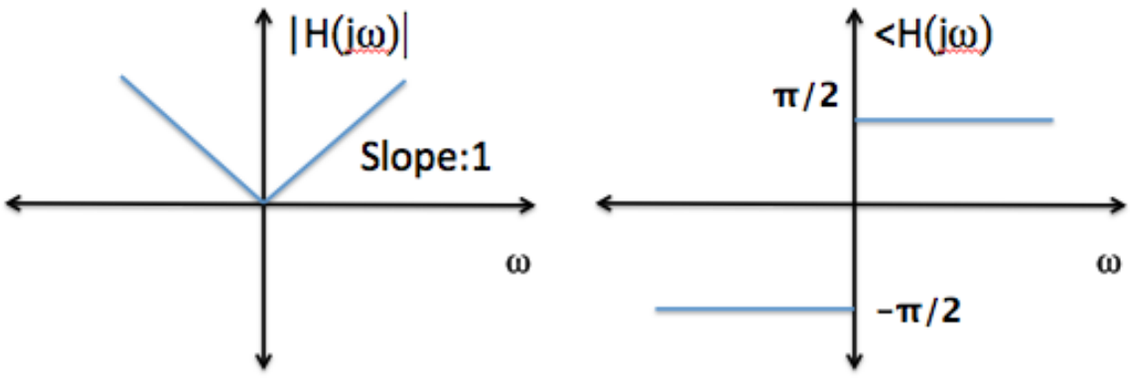


Figure 1: Solution to Qn 6

7. The output of a linear system for a step input is $t^2e^{-t}u(t)$. Find the impulse response, $h(t)$.

   Solution : If $u(t)$ represents the step response, the impulse response is given by, $h(t) = \frac{du}{dt} = (-t^2e^{-t} + 2te^{-t})u(t) = (te^{-t}(2-t))u(t)$

   1 mark to identify that impulse response is the derivative of step response, 1 mark for correct answer. Pl check for alternate solutions.

8. Given that $X(s) = \frac{2s}{s^2+2s+1}$, find the value of $x(t)$ just after $t = 0$.

   Solution : From initial value theorem, $x(t)_{t=0^+} = sH(s)$ evaluated as $s \to \infty$ - (1 mark) $sH(s) = \frac{2s^2}{(s^2+2s+1)} = \frac{2}{(1+\frac{2}{s}+\frac{1}{s^2})} = 2$; when $s \to \infty$. (1 mark).

9. What are the necessary conditions on α , β and n_0 for the system described below to be causal and stable : $y[n] = \alpha y[n-1] + \beta x[n-n_0]$.

   Solution : For x bounded, β should be finite and $-1 \leq \alpha \leq +1$ for stability. For causality, $n_0 \geq 0$.

   1 mark for stability condition, 1 mark for causality conditon.

10. Consider a system with input $x[n]$ and output, $y[n] = cos\left(\frac{\pi}{4}x[n]\right)$ If the input to the system is $\left(\frac{n^2}{2}\right)$ find whether the output is periodic; if periodic, find its fundamental period.

    Solution : $y[n] = cos\left(\frac{\pi}{4}\frac{n^2}{2}\right)$ If $y[n]$ is periodic with period N, $y[n+N] = y[n]$

    $cos\left(\frac{\pi}{4}\frac{(N+n)^2)}{2}\right] = cos\left(\frac{\pi}{4}\frac{n^2}{2}\right)$

    $\frac{\pi}{8}(n+N)^2 = \frac{\pi}{8}(n)^2 \pm 2k\pi$; where k is an integer.

    $N(N+2n) = \pm 16k$

    N has to be an even number since the product $N(N+2n)$ is even.

    Let $N = 2M$, where M is an integer.

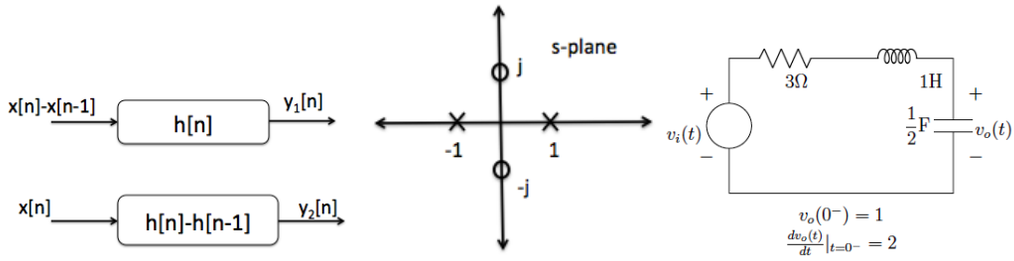
    $i.e, 2M(2M+2n) = 16k$

    or, $M(M+n) = 4k$

    Now the projduct, $M(M+n)$ is even; $M+n$ could be even or odd and hence should correspond to k; and hence M must be equal to 4.

Therefore, the fundamental period, N must be 8.

1 mark for correct procedure, 1 mark for correct answer.



(a) Qn 4 (b) Qn 5 (c) Qn 11

Figure 2: Figures for Questions 4, 5 and 11

11. Determine the differential equation relating $v_i(t)$ and $v_o(t)$ for the RLC circuit shown in Fig 1c. Suppose that $v_i(t) = e^{-3t}u(t)$. Using the unilateral Laplace transform, determine $v_o(t)$ for $t > 0$. (8)

Solution : (a) The differential equation relating $v_i(t)$ and $v_o(t)$ can be expressed as

$$LC\frac{d^2v_o(t)}{dt^2} + RC\frac{dv_o(t)}{dt} + v_o(t) = v_i(t)$$

$$\frac{d^2v_o(t)}{dt^2} + \frac{R}{L}\frac{dv_o(t)}{dt} + \frac{1}{LC}v_o(t) = \frac{1}{LC}v_i(t)$$

Substituting the values of R, L and C, we get

$$\frac{d^2v_o(t)}{dt^2} + 3\frac{dv_o(t)}{dt} + 2v_o(t) = 2v_i(t)$$

(b) Taking the unilateral Laplace transform of the above differential equation, we get

$$s^2V_o(s) - sv_o(0^-) - v_o'(0^-) + 3sV_o(s)$$
$$- 3v_o(0^-) + 2V_o(s) = 2V_i(s)$$

Since, $v_i(t) = e^{-3t}u(t)$,

$$V_i(s) = \frac{1}{s+3}, Re\{s\} > -3$$

Substituting this along with the initial conditions, we get

$$V_o(s) = \frac{(s^2+8s+17)}{(s+1)(s+2)(s+3)}$$

The partial fraction expansion of $V_o(s)$ is

$$V_o(s) = \frac{5}{s+1} - \frac{5}{s+2} + \frac{1}{s+3}$$

Taking the inverse Laplace transform, we get

$$v_o(t) = (5e^{-t} - 5e^{-2t} + e^{-3t})u(t)$$

Marking Scheme : Writing the correct time domain eqn, with values of components subsituted (2)

Writing the correct eqn in LT domain. (2)

Substituting boundary conditions and solving for V(s) (1)

Partial Fraction Expansion (2)

ILT to find V(t) (1)

12. Given two real signals $x(t)$ and $y(t)$.

The convolution of $x(t)$ and $y(t)$ is given by $x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau$.

The correlation between $x(t)$ and $y(t)$ is given by $x(t) \star y(t) = \int_{-\infty}^{\infty} x(\tau)y(\tau-t)d\tau$.

For the $x(t)$ given in the Fig.2a, sketch (i) $x(t) * x(t)$ and (ii) $x(t) \star x(t)$. (4+4=8)

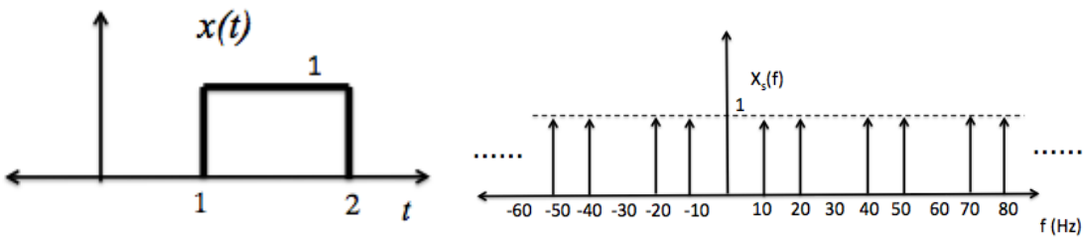
13. Let $x(t) = 60\cos(2\pi f_0 t)$. This signal is sampled at $30Hz$, by multiplying it by an impulse train given as

$\sum_{m=-\infty}^{\infty} \delta(t - \frac{m}{30})$, to obtain the signal, $x_s(t)$.

The corresponding Fourier transform, $X_s(f)$, is shown in Fig 2b.

(a) What is the frequency f_0 of the sinusoid? (2)

(b) Is there a unique solution for f_0 in the above problem? If yes, please justify. If not, find at least two other values. (4)



(a) Qn 12 (b) Qn 13

Figure 3: Figures for Questions 12 and 13

14. Use the properties of Fourier and Laplace transforms to solve the following :

(a) Find the Fourier transform of $x_1(t) = \frac{t}{(1+t^2)^2}$, $t \in \mathbb{R}$. (5)

(b) Find the inverse Laplace transform of $X_2(s) = \ln \frac{s+a}{s+b}$. (Assume that $a, b > 0$ and that $x_2(t)$ is a causal signal). (5)

Solution:

(a) We know that $\frac{1}{1+t^2} \longleftrightarrow \pi e^{-|\omega|}$. (See Example 4.13 O & W). Using the differentiation property, we differentiate in time and write

$$\frac{-2t}{(1+t^2)^2} \longleftrightarrow j\omega\pi e^{-|\omega|}$$

from which

$$\frac{t}{(1+t^2)^2} \longleftrightarrow \frac{-j\omega\pi}{2} e^{-|\omega|}$$

.

(b) From LT property, $\frac{d}{ds}X_2(s) \longleftrightarrow -tx_2(t)$. But, $\frac{d}{ds}X_2(s) = \frac{1}{s+a} - \frac{1}{s+b} \longleftrightarrow \left(e^{-at} - e^{-bt}\right) u(t)$, since the signal is causal. Thus, combining the two, we get

$$x_2(t) = \frac{1}{t}\left(e^{-bt} - e^{-at}\right) u(t).$$

For both these questions, students may choose an alternate method to solve, 2.5 marks for the correct approach, 2.5 marks for the correct answer.

(c) A causal LTI system with impulse response $h(t)$ has the following properties:

(i) When the input is $x(t) = 2e^{2t}$, the output is $y(t) = \frac{1}{3}e^{2t}$.

(ii) The impulse response satisfies $\frac{d}{dt}h(t)+2h(t) = e^{-4t}u(t)+bu(t)$, where b is an unknown constant.

Determine the output of this system, when the input signal is $\cos 3t$. There should be no unknown constants in your answer. (8)

Solution: First observe that (i) implies $H(2) = 1/6$, using the eigenfunction property. Next, taking the BLT of the differential equation, we get $H(s)(s+2) = \frac{1}{s+4} + \frac{b}{s}$. Putting $s = 2$ in the above gives $b = 1$. Thus,

$$H(s) = \frac{2}{s(s+4)}.$$

The response to $\cos 3t$ again using the eigenfunction property, is $\frac{H(3j)e^{3jt}+H(-3j)e^{-3jt}}{2} = -\frac{je^{3jt}}{3(4+3j)} + \frac{je^{-3jt}}{3(4-3j)}$.

2 marks for finding H(2); 3 marks for finding b; 3 marks for finding the output of cos(3t)

(d) Consider an ideal lowpass filter with cutoff frequency 500 Hz as an LTI system H_l whose frequency response is given in Fig 3a. (3+7=10)

i. What is the response of this filter to the input signal $x(t) = \cos(1500\pi t)+\sin(3000\pi t)$?

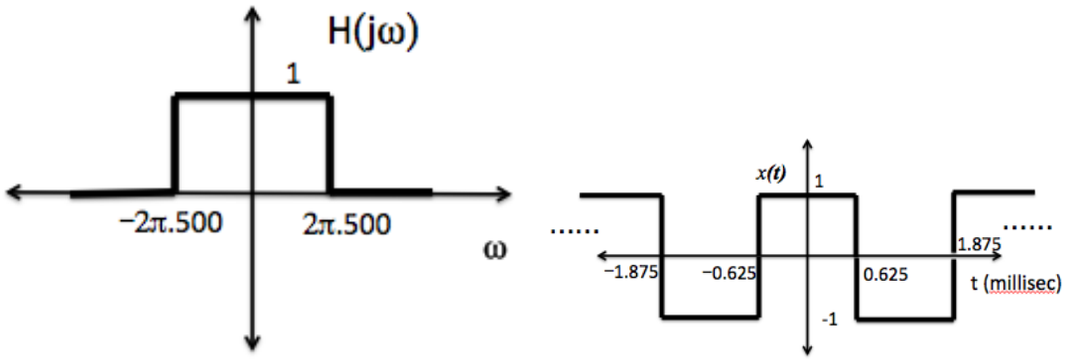
ii. What is the output $y(t)$ of this filter for an input, $x(t)$, which is a periodic square wave with period 2.5 ms, as shown in Fig 3b.

(e) The impulse response $h(t)$ of a causal continuous-time LTI system is known to be real-valued. Due to some practical constraints, suppose only the *real part* of the frequency response of the system is available to us.

i. Express the inverse Fourier transform of $\mathrm{Re}\{H(j\omega)\}$ in terms of $h(t)$.

ii. From your answer to the previous part, show how we can determine $h(t)$ completely, although only the real part of the frequency response is available to us. (10)

i. $\mathrm{Re}\{H(j\omega)\} = \frac{H(j\omega)+H^*(j\omega)}{2} = \frac{H(j\omega)+H(-j\omega)}{2}$, since $h(t)$ is real. Next, note that $H(-j\omega) \longleftrightarrow h(-t)$. Thus it follows that $\mathrm{Re}\{H(j\omega)\} \longleftrightarrow \frac{h(t)+h(-t)}{2}$, so that the IFT of $\mathrm{Re}\{H(j\omega)\}$ is simply the even part of $h(t)$.



(a) (b)

Figure 4: Figures for Questions 16

ii. $h(t)$ can be recovered from the IFT of $\mathrm{Re}\{H(j\omega)\}$ by multiplying the result (in the previous part) by $2u(t)$. To see this, note that since the system is causal, the impulse response satisfies $h(t)u(t) = h(t)$ and $h(-t)u(t) = 0$, for all t. Thus, to recover $h(t)$, we take the IFT of $\mathrm{Re}\{H(j\omega)\}$, and multiply the result by $2u(t)$.

End

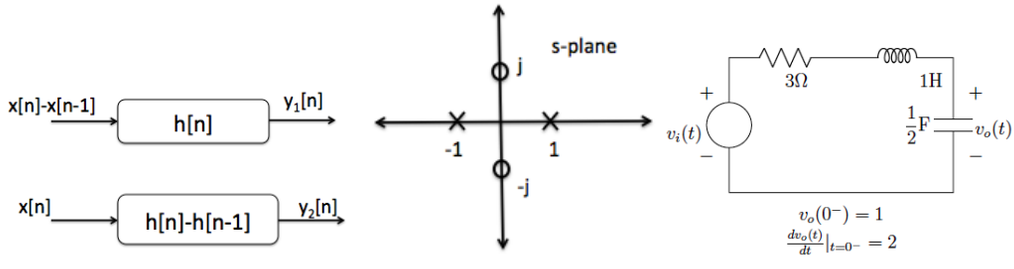
End Semester Examination
EE1101 Signals and Systems - Jan-May 2018

Duration:3 Hrs **Max Marks : 80**

Instructions

- ***Qn 1-10** : **2 marks each**. Answer briefly, with appropriate justification/counter example(s).*
- ***Qn 11-17** - Marks are shown in brackets. Show all working steps neatly.*
- *Do not write anything in the question paper. Attach the question paper with your answersheet.*

1. Does a two-sided signal exist such that the RoC of its Laplace Transform is a left half-plane ?
2. Find the RoC of the Laplace transform of the signal $x(t) = \cos \pi t$.
3. When a causal LTI system is cascaded with a non-causal LTI system, then the resultant system must be non-causal. Is this statement true or false ?
4. For the LTI systems shown in Fig.1a., $y_1[n] = y_2[n]$. Is this statement true or false ?
5. Find whether pole-zero plot shown in Fig. 1b could correspond to an even function in time ? If yes- find the required ROC. (Given that the transfer function is rational.)
6. Sketch the magintude and phase response ($H(j\omega)$) of a system that performs differentiation.
7. The output of a linear system for a step input is $t^2e^{-t}u(t)$. Find the impulse response, $h(t)$.
8. Given that $X(s) = \frac{2s}{s^2+2s+1}$, find the value of $x(t)$ just after $t = 0$.
9. What are the necessary conditions on α , β and n_0 for the system described below to be causal and stable : $y[n] = \alpha y[n-1] + \beta x[n-n_0]$.
10. Consider a system with input $x[n]$ and output, $y[n] = cos\left(\frac{\pi}{4}x[n]\right)$ If the input to the system is $\left(\frac{n^2}{2}\right)$ find whether the output is periodic; if periodic, find its fundamental period.



(a) Qn 4 (b) Qn 5 (c) Qn 11

Figure 1: Figures for Questions 4, 5 and 11

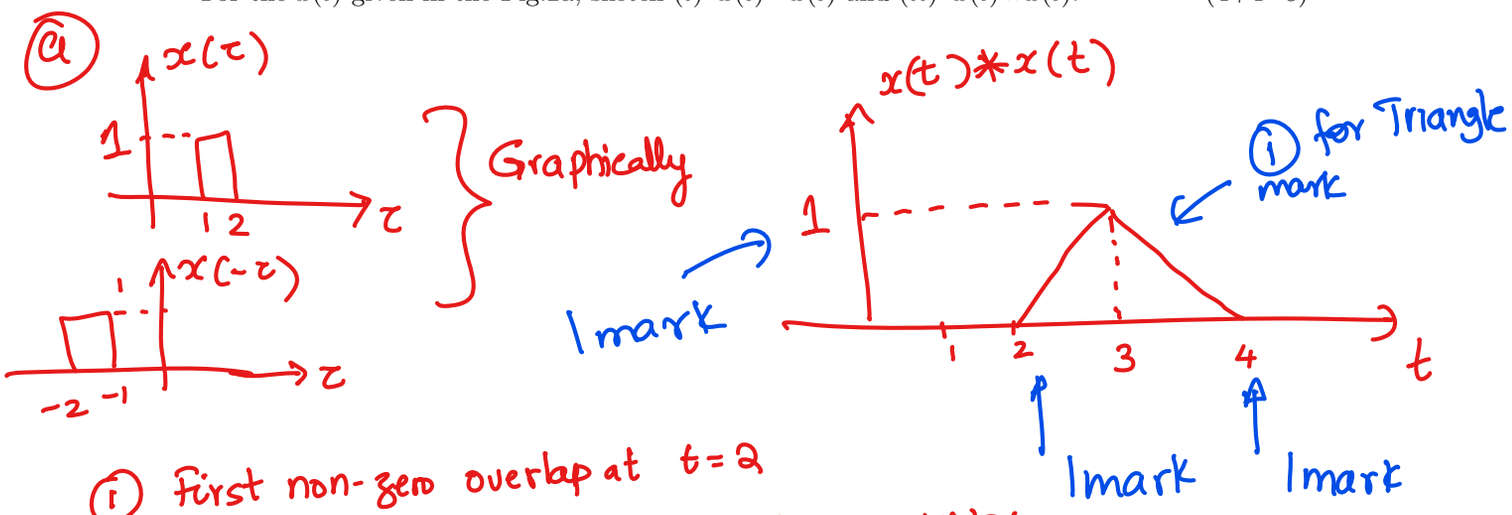
11. Determine the differential equation relating $v_i(t)$ and $v_o(t)$ for the RLC circuit shown in Fig 1c. Suppose that $v_i(t) = e^{-3t}u(t)$. Using the unilateral Laplace transform, determine $v_o(t)$ for $t > 0$. (8)

12. Given two real signals $x(t)$ and $y(t)$.

    The convolution of $x(t)$ and $y(t)$ is given by $x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau$.

    The correlation between $x(t)$ and $y(t)$ is given by $x(t) \star y(t) = \int_{-\infty}^{\infty} x(\tau)y(\tau-t)d\tau$.

    For the $x(t)$ given in the Fig.2a, sketch (i) $x(t) * x(t)$ and (ii) $x(t) \star x(t)$. (4+4=8)

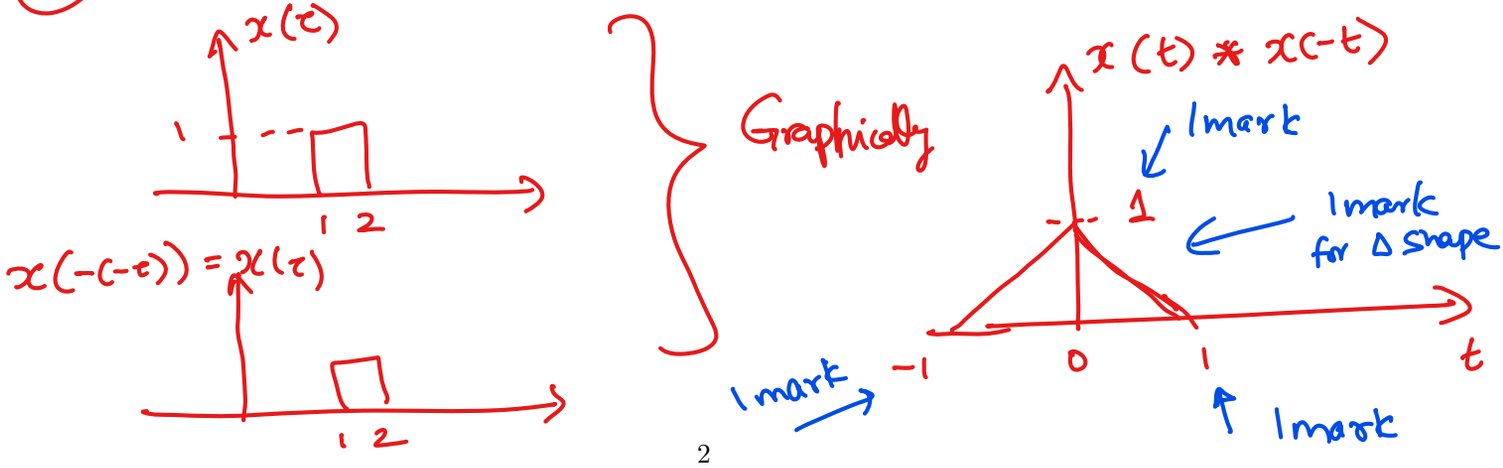


① First non-zero overlap at $t=2$

② Complete overlap $t=3 \Rightarrow$ Area $= 1\times1=1$

③ Final overlap at $t=4$

b Correlation: $x(t) * x(-t) = x(t) \star x(t)$ (Conv. / Correlation)



① Complete overlap at $t=0$ } Area of overlap $= 1\times1=1$

② To the left last overlap $t=-1$

③ To the right last overlap $=+1$

13. Let $x(t) = 2\cos(2\pi f_0 t)$. This signal is sampled at $30Hz$, by multiplying it by an impulse train given as

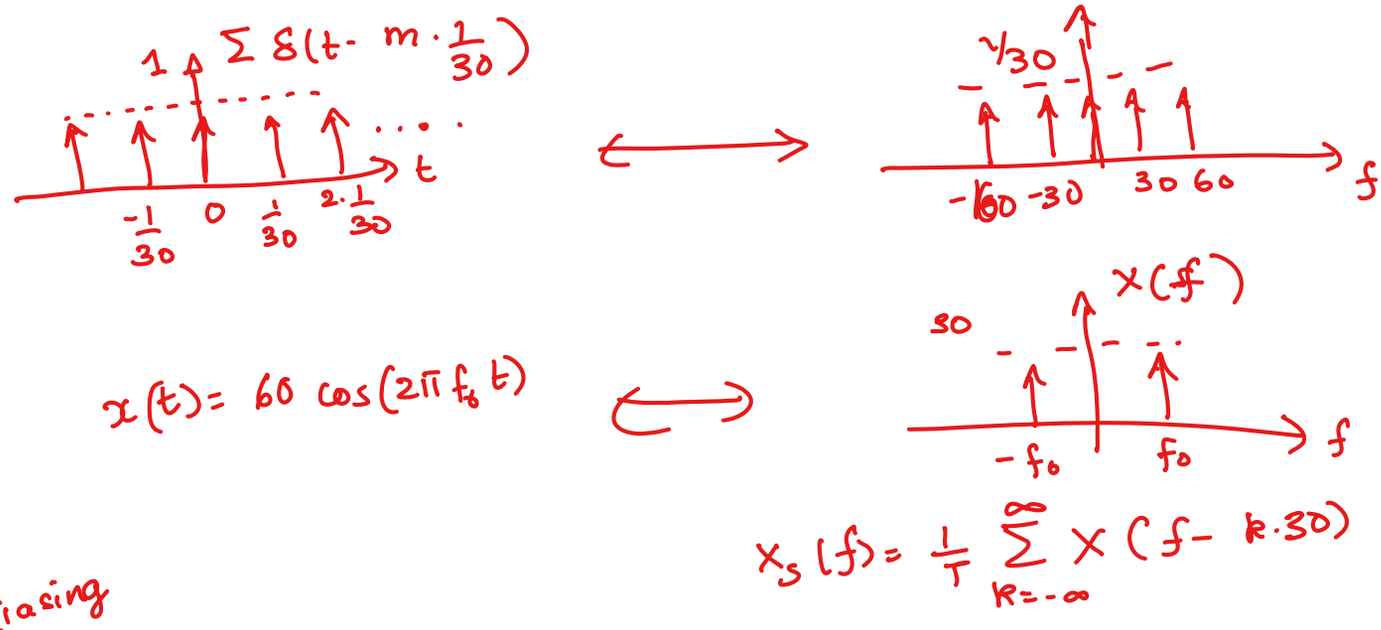
$\sum_{m=-\infty}^{\infty} \delta(t - \frac{m}{30})$, to obtain the signal, $x_s(t)$.

The corresponding Fourier transform, $X_s(f)$, is shown in Fig 2b.

(a) What is the frequency f_0 of the sinusoid? (2)

(b) Is there a unique solution for f_0 in the above problem? If yes, please justify. If not, find at least two other values. (4)

60



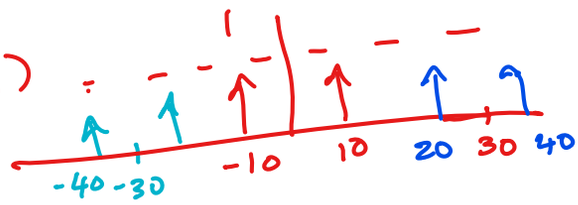
$$X_s(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - k \cdot 30)$$

No Aliasing

① If $f_0 = 10$ Hz

2 marks

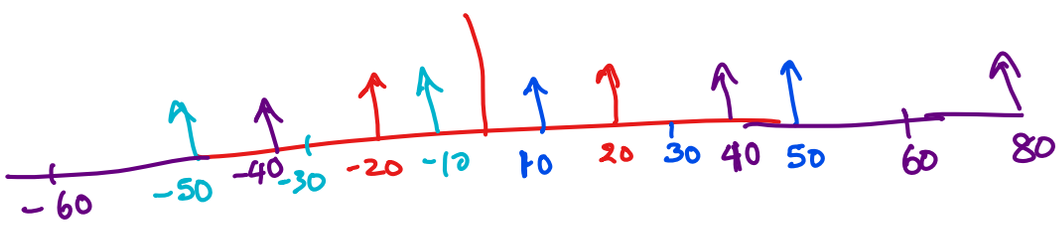
$\Rightarrow X_s(f)$



Aliasing

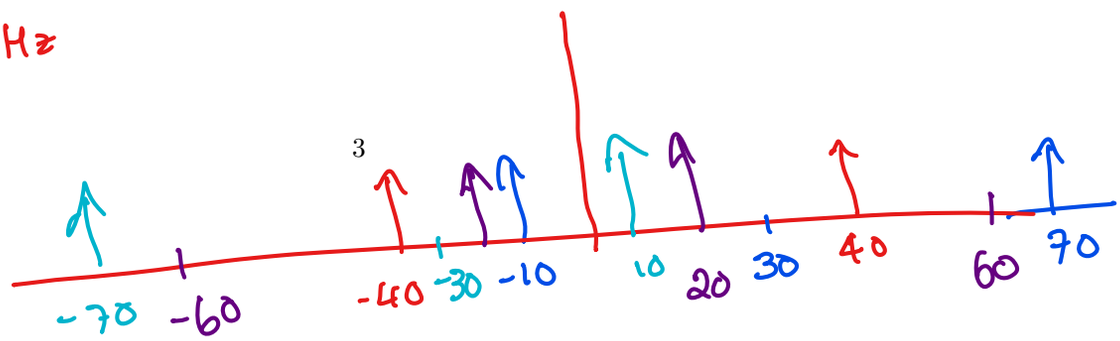
② If $f_0 = 20$ Hz

2 marks



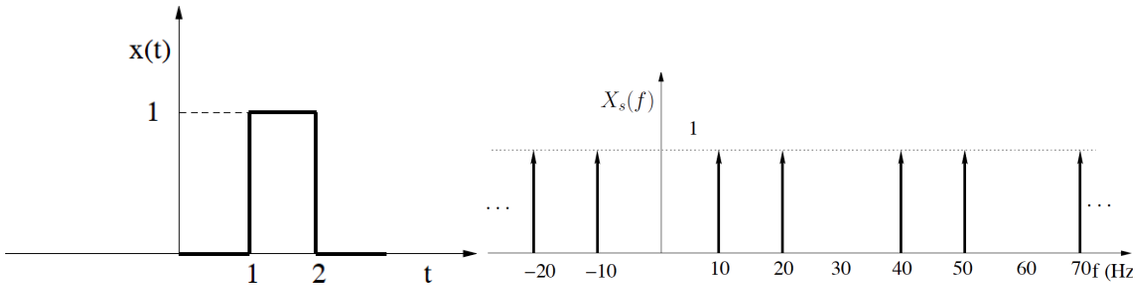
③ If $f_0 = 40$ Hz

2 marks



Similarly $f_0 = 50$Hz is also okay

Note : Give marks only if they explain using the figures or $\frac{1}{T}\sum_{k=-\infty}^{\infty} X(f - k \cdot 30)$



(a) Qn 12 (b) Qn 13

Figure 2: Figures for Questions 12 and 13

14. Use the properties of Fourier and Laplace transforms to solve the following :

    (a) Find the Fourier transform of $x_1(t) = \frac{t}{(1+t^2)^2}$, $t \in \mathbb{R}$. (5)

    (b) Find the inverse Laplace transform of $X_2(s) = \ln \frac{s+a}{s+b}$. (Assume that $a, b > 0$ and that $x_2(t)$ is a causal signal). (5)

15. A causal LTI system with impulse response $h(t)$ has the following properties:

    (i) When the input is $x(t) = 2e^{2t}$, the output is $y(t) = \frac{1}{3}e^{2t}$.

    (ii) The impulse response satisfies $\frac{d}{dt}h(t) + 2h(t) = e^{-4t}u(t) + bu(t)$, where b is an unknown constant.

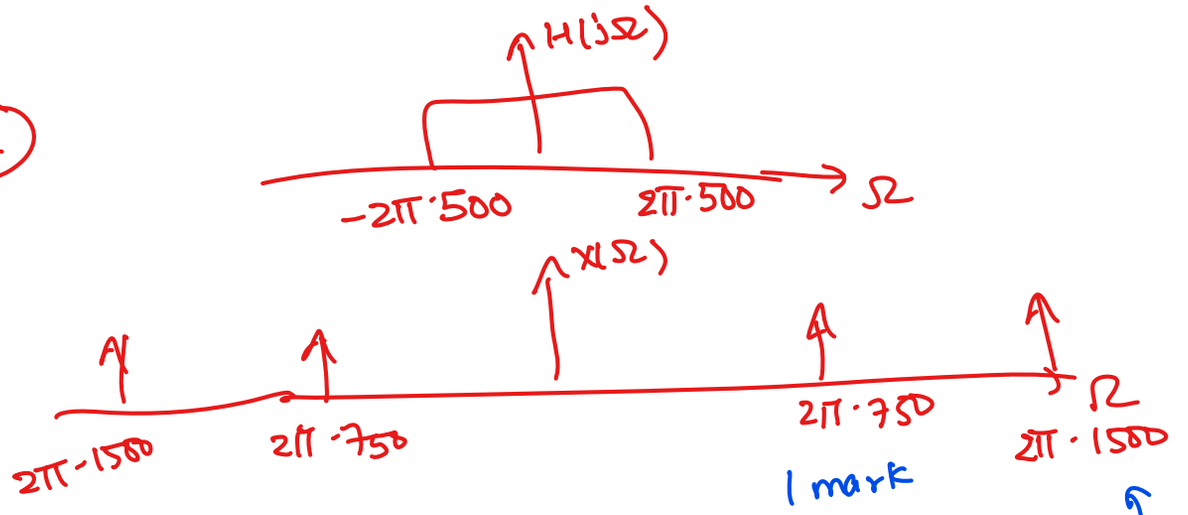
    Determine the output of this system, when the input signal is $\cos 3t$. There should be no unknown constants in your answer. (8)

16. Consider an ideal lowpass filter with cutoff frequency 500 Hz as an LTI system H_l whose frequency response is given in Fig 3a. (3+7=10)

    (a) What is the response of this filter to the input signal $x(t) = \cos(1500\pi t) + \sin(3000\pi t)$?

    (b) What is the output $y(t)$ of this filter for an input, $x(t)$, which is a periodic square wave with period 2.5 ms, as shown in Fig 3b.

(a)



The response is zero ⇒ 1 mark

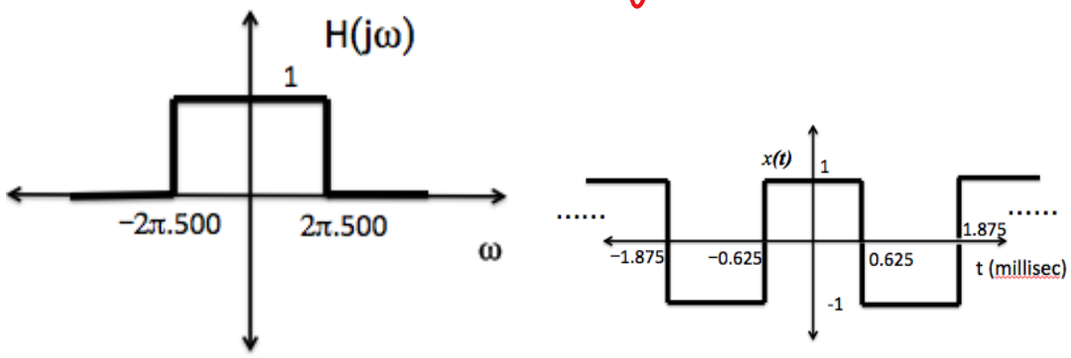
(b) Note over one period area of $x(t) = 0$

$\Rightarrow a_0 = \frac{1}{T}\int_T x(t)\,dt = 0$ ⇒ 2 mark

secondly:- $T = 2.5\ ms \Rightarrow F_0 = \frac{1}{2.5\times10^{-3}} = 400\ Hz$ (1) mark

only a_1, a_{-1} $\Rightarrow$ Component will pass through the low pass filter (1) mark

$\sum a_k e^{jk.2\pi f_0 t}$



(a) (b)

Figure 3: Figures for Questions 16

17. The impulse response $h(t)$ of a causal continuous-time LTI system is known to be real-valued. Due to some practical constraints, suppose only the *real part* of the frequency response of the system is available to us.

    (a) Express the inverse Fourier transform of $\mathrm{Re}\{H(j\omega)\}$ in terms of $h(t)$.

    (b) From your answer to the previous part, show how we can determine $h(t)$ completely, although only the real part of the frequency response is available to us. (10)

End

(b) Hence $a_{-1}e^{-j1.2\pi F_0 t} + a_1 e^{j1.2\pi F_0 t}$

But since $x(t)$ is real and even

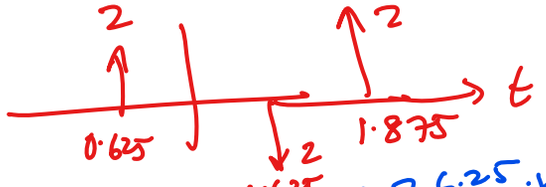
$\Rightarrow a_1 = a_{-1}$ $\Rightarrow$ real and even

2 mark

$y(t) = 2a_1 \cos(2\pi f_0 t) = 2a_1 \cos(2\pi \cdot 400 t)$

1 mark $a_1 = \frac{1}{2.5}\left[\int_{-0.625}^{0.625} 1.\ e^{j1.2\pi f_0 t} + \int_{0.625}^{1.875}(-1)\ e^{j\cdot1.2\pi f_0 t}\right]$

$\Rightarrow$ Easier way $g(t) = \frac{d}{dt}x(t)$ $\leftrightarrow$ $b_k = \frac{2}{2.5}\left[e^{jk\frac{2\pi}{2.5}0.625} - e^{jk\frac{2\pi}{2.5}0.625}\right]$



$\Rightarrow a_1 = \frac{1}{j2\pi\cdot400} b_1$

Subjective type questions [30]

1. The response of an LTI system to the input $x_1(t) = 1,\ t \in [0,1]$, (and zero otherwise) is given by $y_1(t) = (1-e^{-t})u(t) - (1-e^{-(t-1)})u(t-1)$. Determine the output $y_2(t)$ of the system, when the input is as given below [4].

$$x_2(t) = \begin{cases} t,\ t \in [0,1], \\ 2-t,\ t \in [1,2], \\ 0 \text{ otherwise} \end{cases}.$$

2. State whether the following statements are true or false, with justification or counterexample [10].

   (i) If the region of convergence (RoC) of the transfer function of an LTI system is a right half plane, then the system is causal.

   (ii) If the RoC of $X(s)$ is a right half plane, then $x(t)$ is a right-sided signal.

   (iii) If the RoC of a transfer function of an LTI system includes the origin, then the system is BIBO-stable.

   (iv) If the frequency response $H(j\omega)$ of an LTI system is well-defined and finite for all $\omega \in \mathbb{R}$, then the system is BIBO stable.

   (v) If $x(t)$ is an even signal, then the ROC of $X(s)$ is symmetric about the $j\omega$ axis.

3. Find the Laplace transform, or the inverse Laplace transform of the following (a and b are positive, real): [4]

   (i) $x_1(t) = te^{-\frac{t^2}{2}},\ t \in \mathbb{R}$.

   (ii) $X_3(s) = \ln[(s+a)^2 + b^2]$; $x_3(t)$ is a causal signal.

4. Find the transfer function relating $Y(s)$ to $X(s)$ corresponding to the following block diagram [2]:

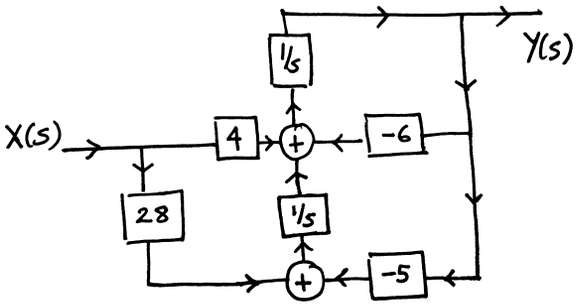


Figure 1: Block diagram

5. Consider a real bandpass signal, which has non-zero frequency components only in the range $|\omega| \in (\frac{\omega_0}{2}, \omega_0]$, where $\omega_0 > 0$. We are interested in perfectly recovering this signal, from its samples.

   (i) What is the minimum sampling rate required, if we have access only to an ideal low-pass filter? Specify the cut-off frequencies of the filter. [2]

   (ii) What is the minimum sampling rate, if we have access to an ideal bandpass filter? Specify the cut-off frequencies of the filter. [3]

6. Consider an RLC circuit, where the three components are connected in series. The inductor value L and the capacitor value C are given constants, but the resistance R can be changed. At time $t = 0^-$, suppose that the capacitor voltage and inductor current are both non-zero. For what values of R does the voltage across the capacitor behave monotonically for $t > 0$? [5]

EE1101 – Signals and Systems – Jan - May 2017

End Semester Examination

Name:	**Section**: KJ | UK | DV | VV **Roll**:

Instructions; Use pen only; For the objective type questions, answer on the question paper itself; Return the question paper along with your answer script.

1 Objective type questions [20]

1. Starting with $f(t)$, we are to arrive at $f(at-b)$ by shifting and scaling. We can either scale by _____ and then shift by _____, or shift by _____ and then scale by _____. [2]

2. A function $f(t)$ is non-zero only on $t = [2, 3)$, linearly decreasing from 1 at $t = 2$ to 0 at $t = 3$. Express this function in terms of functions $t, u(t)$ or their shifted/scaled versions: _____ [2]

3. Given that a smooth function $\phi(t)$ has $\phi(0) = 2$, the value of $\int_{-\infty}^{-1} \phi(t)\delta(t)dt$ is _____. [1]

4. The function $f(t) = \alpha \exp(-\alpha t)$ approaches a dirac delta function $\delta(t)$, if (circle the correct answer(s)): (a) $\alpha \to 0$, (b) $\alpha \to \infty$, (c) $t \to \infty$, (d) none of the above. [1]

5. They system $y(t) = \exp(x(t))$ is BIBO stable/unstable, and the output to an input $x(t) = tu(t)$ is bounded/unbounded. (Circle the correct answers) [1]

6. The discrete signal $x[n] = cos(2\pi fn)$ is periodic if f is _____. [1]

7. For a memoryless LTI system, the ratio between $y(t)$ and $x(t)$ is: (a) constant, (b) time-dependent, (c) requires more information to be provided, (d) $\delta(t)$. (circle the correct answer(s))[1]

8. For a complex Fourier series to represent a real signal, the condition on the Fourier series coefficient is _____. [1]

9. The periodic function $x(t) = \ln(t)$ defined over a period $0 < t \leq 1$, does not have Fourier series representation because it does not satisfy the specific condition of _____. [1]

10. If an audio signal is connected across a series R-C circuit and the output is taken across the resistor, which frequencies are enhanced?
    (a) Low, (b) High, (c) All equally enhanced, (d) All equally suppressed. [1]

11. If $X(j\omega) \longleftrightarrow x(t)$ denotes a Fourier pair, then complete the following Fourier pair: $\frac{dX(j(\omega-\omega_o))}{d\omega} \longleftrightarrow$ _____. [1]

12. The ideal low pass filter with cut-off ω_c has impulse response $h(t) =$ _____. This is not realizable because _____. [2]

13. For $x(t) = -\exp(-at)u(-t)$ and $a > 0$, the Fourier transform of $x(t)$ exists/doesn't exist (circle the correct answer). If it exists, it is _____, and if not, this is why: _____ [2].

14. Given a transfer function $H(s) = s/[(s+a)(s+b)]$, with $a > 0 > b$, of the three possible ROCs for this function, which of them is causal? _____. [1]

15. In solving a nth order ordinary differential equation initial value problem, the number of initial conditions required is: (a) 1, (b) n, (c) $n-1$, (d) n^2. [1]

16. Consider the bilateral Laplace transform of $x(t)$ to be $X(s)$, which is known to be a rational function, with poles and zeroes strictly in the left half plane. The final value of such a function, $x(\infty)$, is _____. [1]
